



Ahsanullah University of Science and Technology
Department of Computer Science and Engineering

PROJECT PROPOSAL REPORT

Explainable Fake Review Detection for Bangla

A Scoped, Explainable, Multi-Signal NLP Pipeline with
High-Confidence Pseudo-Labeling

Course No.: CSE 4262

Course Name: Data Analytics Lab

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Abstract

Online reviews influence purchasing decisions, yet promotional, copied, or otherwise deceptive reviews can distort that evidence. Detecting such content in Bangla is difficult because annotated resources are limited, informal writing is frequently code-mixed, and many high-performing published systems depend on computationally expensive transformer ensembles. This proposal presents a resource-conscious and explainable fake-review detection pipeline implemented primarily with PySpark, Spark SQL, and Spark MLlib. The Bengali Fake Review Detection (BFRD) dataset will provide 9,049 gold-labeled food reviews: 1,339 fake and 7,710 non-fake. BanglishRev will supply a large auxiliary pool of Bangla, English, Banglish, and code-mixed e-commerce reviews without treating their ratings or sentiment labels as evidence of authenticity. A classifier trained only on the BFRD training partition will assign pseudo-labels to this pool, and only predictions with confidence strictly greater than 0.90 will enter the expanded training set. An untouched, gold-labeled BFRD test partition will remain isolated throughout the process.

The proposed feature space will combine TF-IDF, Word2Vec document representations, stylometric and promotional signals, and near-duplicate features derived through MinHash locality-sensitive hashing. Logistic Regression, Naive Bayes, Random Forest, Gradient-Boosted Trees, and Linear Support Vector Classification will be compared under gold-only and semi-supervised settings. Macro-F1 and precision-recall area under the curve will receive priority over accuracy because the gold data are strongly imbalanced. Spark SQL will support auditing, provenance, split integrity, pseudo-label filtering, and experiment summaries, while model coefficients, tree-based feature importance, and a small LIME case study will support explanation. The central experiment will determine whether carefully filtered, in-domain auxiliary data improve generalization without contaminating evaluation or concealing pseudo-label noise.

Keywords: Bangla NLP, fake review detection, PySpark, Spark SQL, semi-supervised learning, pseudo-labeling, TF-IDF, Word2Vec, MinHash LSH, explainable machine learning

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1 Project Title

Explainable Fake Review Detection for Bangla

A PySpark and Spark SQL pipeline using multi-signal text features, near-duplicate analysis, and high-confidence pseudo-labeling

The title reflects the project’s principal task and intended contribution. The system will classify Bangla reviews as fake or genuine while retaining interpretable evidence from lexical, semantic, stylometric, promotional, and text-similarity signals. PySpark will provide the distributed data-processing and machine-learning framework; it is an implementation technology rather than part of the project name.

2 Group Information

Table 1: Proposal and group information

Item	Information
Project title	Explainable Fake Review Detection for Bangla
Group number	3
Lab group / section	4
Students	Syed Abir Hossain (20220104013) Alim Bin Yeasin (20220104016) Shuhrid Abrar (20220104028)
Course	CSE 4262: Data Analytics Lab
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Submission date	25 August 2026
Primary dataset	Bengali Fake Review Dataset (BFRD) [1, 2]
Auxiliary dataset	BanglishRev e-commerce review dataset [3, 4]

Both datasets are publicly accessible for research under their respective repository licenses. Direct links and complete citations are provided in the References section.

3 Introduction

Online reviews influence purchasing decisions, brand reputation, and the visibility of products and services. Their value depends on authenticity: promotional, copied, or deliberately deceptive reviews can distort the evidence available to customers and businesses. Automatic detection is therefore useful for marketplace moderation, consumer protection, and data-quality analysis.

The problem is particularly challenging in Bangla. Reviews frequently contain spelling variation, informal punctuation, emoji, English expressions, Romanized Bangla, and code-mixed language. Bangla also has fewer labeled resources and mature preprocessing tools than English. A reliable pipeline must therefore look beyond individual words and combine several types of evidence, including textual content, writing style, promotional patterns, and repeated or templated language.

The BFRD benchmark provides gold supervision for this task through 9,049 expert-annotated food reviews, consisting of 1,339 fake and 7,710 non-fake instances [1]. Its published work evaluates deep neural and transformer-based systems. This project will instead establish a scalable, resource-conscious, and explainable classical machine-learning pipeline in PySpark. BanglishRev will supply a much larger pool of Bangla, English, Banglish, and code-mixed e-commerce reviews [3]. Because BanglishRev has no fake-review ground truth, it will be incorporated only through controlled

pseudo-labeling.

Consumers, e-commerce platforms, review moderators, and Bangla NLP researchers may benefit from the analysis. Spark SQL will support reproducible auditing and aggregation, while Spark MLlib will enable distributed text features, similarity search, classification, and model comparison. The project will consequently demonstrate both a practical application of data analytics and a careful experimental study of whether unlabeled data improve detection.

3.1 Research Context

Earlier opinion-spam research established supervised lexical baselines and demonstrated that deceptive writing can contain measurable linguistic patterns [5, 6]. BFRD extended this task to Bengali and used augmentation, transformer ensembles, and LIME explanations [1]. The proposed work will complement that research by focusing on Spark-native classical models, explicit feature provenance, near-duplicate signals, and a gold-only versus pseudo-labeled ablation.

4 Problem Statement

Bangla fake-review detection has three connected limitations. First, the only gold dataset selected for this project is small and strongly imbalanced: fake reviews constitute approximately 14.8% of BFRD. A model can therefore obtain misleadingly high accuracy while failing to identify the minority fake class. Second, BFRD is restricted to food-related reviews collected from social media, so a model trained on it may not generalize to broader e-commerce language. Third, text alone can be ambiguous; genuine and deceptive reviews may share sentiment and vocabulary, while duplicated templates, promotional markers, or unusual writing patterns may provide additional evidence.

BanglishRev offers scale and language diversity but does not label review authenticity. Its ratings indicate sentiment, not whether a review is fake. Treating those ratings as authenticity labels would invalidate the task. The missing information is therefore trustworthy fake/genuine supervision for the auxiliary corpus and evidence about whether high-confidence model-generated labels can improve performance without introducing excessive noise or source bias.

This project will address the problem through a leakage-controlled semi-supervised experiment. A classifier trained only on the gold BFRD training partition will score BanglishRev reviews. Only predictions with confidence strictly greater than 0.90 will be accepted as pseudo-labels, and those records will enter training only. The final decision will be based on comparison against an untouched gold BFRD test set using imbalance-sensitive metrics. Data analytics will support this decision by measuring class distributions, language patterns, duplicate clusters, pseudo-label acceptance, model errors, and the contribution of each feature family.

Central research question: Will a multi-signal PySpark classifier trained with carefully filtered BanglishRev pseudo-labels achieve better macro-F1 and PR-AUC on unseen gold BFRD reviews than an equivalent model trained on BFRD alone?

5 Project Objectives

The project will pursue five specific and measurable objectives:

1. To acquire, audit, deduplicate, and standardize the BFRD and BanglishRev text data in Spark DataFrames while preserving source and label provenance.
2. To construct TF-IDF, Word2Vec, stylometric/promotional, and near-duplicate feature groups using PySpark, Spark SQL, and MinHash locality-sensitive hashing.
3. To train and compare Logistic Regression, Naive Bayes, Linear Support Vector Classification, Random Forest, and Gradient-Boosted Tree classifiers under class imbalance.
4. To generate BanglishRev pseudo-labels at confidence > 0.90 and quantify their effect through a

controlled BFRD-only versus BFRD-plus-pseudo-label ablation.

5. To evaluate and explain the selected model using macro-F1, PR-AUC, class-wise metrics, confusion matrices, feature importance or coefficients, and a small LIME case study.

Achievement of the objectives will be demonstrated through reproducible row audits, saved feature/model pipelines, comparison tables, visualizations, and a final report. Improvement from pseudo-labeling will be treated as an experimental result rather than a predetermined outcome.

6 Dataset Description

Table 2: Datasets proposed for the study

Dataset	Confirmed source facts	Proposed use
BFRD	9,049 rows in two CSV files; two principal columns, Review and Label ; 1,339 fake (0) and 7,710 non-fake (1).	Gold source for supervised training, validation, and final testing [1, 2].
BanglishRev	1.74 million written reviews associated with 3.2 million ratings and about 128,000 products; JSON review/metadata files and separate image archives; no fake-review label.	Unlabeled text pool for pseudo-labeling; image content will be excluded [3, 4].

6.1 Availability and Target Variable

BFRD is the primary dataset because its target variable directly represents the task: **Label=0** denotes fake and **Label=1** denotes non-fake. The source contains food-related Bangla reviews collected from public social-media content and annotated by experts [1]. The project will verify all published row counts and column names after download rather than assuming that repository files are unchanged.

BanglishRev is a public large-scale e-commerce resource containing Bengali, English, mixed-language, and Romanized Bengali reviews together with metadata such as rating, review/purchase dates, engagement, seller response, and associated images [3]. Those fields do not define authenticity. The project will use only the written-review and relevant non-identifying metadata portions that can be processed within available resources.

Dataset links:

- [BFRD repository on Hugging Face](#)
- [BanglishRev repository on Hugging Face](#)

6.2 Split and Label Policy

BFRD will be deduplicated and assigned with a fixed seed to stratified gold training, validation, and test partitions of approximately 70%, 15%, and 15%. Duplicate groups will remain within one partition. The gold test partition will be locked before feature fitting, model selection, or pseudo-labeling. Accepted BanglishRev pseudo-labels will be marked explicitly and will never enter validation or testing.

7 Data Attributes

Table 3 summarizes the important source variables. Exact BanglishRev field names will be confirmed during ingestion because repository versions may package metadata differently.

Table 3: Important source and derived attributes

Attribute	Origin	Description and analytical role
Review / review text	Both	Raw textual content; the principal input for preprocessing and feature construction.
Label	BFRD	Gold target: 0=fake and 1=non-fake.
Rating and product metadata	BanglishRev	Retained for source auditing and EDA; not converted into authenticity labels.
Dates and engagement fields	BanglishRev	Optional descriptive metadata, subject to availability and privacy review.
review_id	Generated	Stable identifier used for joins, deduplication, and similarity features.
label_source	Derived	Distinguishes gold , pseudo , and unlabeled records.
confidence_score	Derived	Probability assigned during pseudo-labeling; excluded from predictive features.
source_dataset / source_split	Derived	Records provenance and the immutable train/validation/test assignment.

The feature table will add clean text, token arrays, TF-IDF vectors, Word2Vec representations, and compact engineered variables. The latter will include text length, token count, code-mix ratio, emoji/hashtag/exclamation counts, URL/phone/price flags, promotional-keyword hits, repetition ratio, exact-duplicate status, graph degree, and duplicate-cluster size. Raw text and derived features will remain separate so that transformations can be audited.

8 Proposed Data Analytics Tasks

8.1 Data Collection and Loading

The BFRD CSV files and the textual portion of BanglishRev will be downloaded from their cited Hugging Face repositories. PySpark will load each source with an explicit schema; raw row counts, null counts, file hashes, and source versions will be recorded. Spark SQL views will standardize field names and produce a unified provenance-aware table. Parquet will be used for cleaned intermediate data because it preserves schema and supports efficient repeated analysis.

8.2 Data Preprocessing

Preprocessing will include the following tasks:

- remove or quarantine blank text, invalid labels, corrupt rows, and exact duplicate identifiers;
- normalize Unicode, whitespace, control characters, and selected punctuation without destroying meaningful Bangla text;
- count emoji, hashtags, exclamation marks, URLs, telephone numbers, and price expressions before stripping them from modeling text;
- tokenize with a Bangla-aware library through a batched pandas UDF and remove versioned Bangla/English stop words;
- measure Bangla, Latin, numeric, and code-mixed character composition;
- keep duplicate groups within one data partition to reduce train-test leakage; and
- compute class weights and check extreme text lengths rather than removing outliers automatically.

Continuous handcrafted variables will be scaled only for models that benefit from scaling. Categorical provenance variables will remain audit fields and will not be allowed to become shortcuts for the target.

8.3 Exploratory Data Analysis

EDA will be performed before final modeling and separately for gold and unlabeled sources. Planned outputs include:

- class-frequency and class-percentage charts for BFRD;
- text-length and token-count histograms and box plots by gold class;
- bar charts for emoji, URL, price, telephone, and promotional-keyword signals;
- distributions of Bangla/Latin code-mixing ratios;
- exact- and near-duplicate cluster-size distributions;
- group-wise comparisons of feature means/medians between fake and genuine gold reviews; and
- pseudo-label confidence and acceptance counts by predicted class and source subset.

Spark SQL aggregations will generate the analytical tables, while Matplotlib or Seaborn will render sampled or aggregated outputs. Correlations among compact numeric features will be examined, but sparse TF-IDF columns will not be converted into an impractically dense correlation matrix.

8.4 Feature Engineering and Modeling

Four feature tracks will be constructed. TF-IDF will provide sparse lexical evidence through training-fitted `CountVectorizer` and IDF. Spark MLlib `Word2Vec` will produce dense semantic review vectors. Handcrafted features will represent style, promotion, repetition, length, and language mixing. Finally, token shingles will be converted to binary sparse vectors for `MinHashLSH`; thresholded Jaccard-neighbor pairs will generate duplicate-cluster indicators. Binary shingles are used because `MinHash` operates on set membership rather than TF-IDF weights.

Table 4: Planned classification models and feature settings

Model	Primary features	Purpose
Logistic Regression	TF-IDF + engineered features	Interpretable probability baseline and pseudo-label generator.
Naive Bayes	Non-negative TF-IDF/count features	Efficient classical text baseline.
Linear SVC	Sparse TF-IDF + engineered features	Strong margin-based baseline for high-dimensional text.
Random Forest	Word2Vec + engineered/graph features	Non-linear interactions and feature importance.
Gradient-Boosted Trees	Word2Vec + engineered/graph features	Sequential tree ensemble on compact dense features.

Logistic Regression and one tree ensemble will receive focused hyperparameter tuning through `ParamGridBuilder` and cross-validation. Instance weights will address class imbalance where supported. Naive Bayes will receive only non-negative inputs, and tree models will avoid the very high-dimensional sparse TF-IDF representation.

8.5 Model Evaluation

The locked gold test set will be evaluated only after feature and model choices are fixed. Macro-F1 and precision-recall area under the curve (PR-AUC) will be primary because BFRD is imbalanced. Accuracy, macro precision/recall, ROC-AUC, fake-class precision/recall/F1, and confusion matrices will provide supporting evidence. Where probability output is available, validation-set reliability analysis will check whether confidence is sufficiently meaningful for pseudo-labeling [8].

The central ablation will compare identical model/feature configurations trained on (A) BFRD gold data only and (B) BFRD plus accepted BanglishRev pseudo-labels. Additional ablations will remove handcrafted or near-duplicate features. Error analysis will group false positives and false negatives by length, code mixing, promotional signals, and duplicate membership.

9 Proposed Methodology

9.1 End-to-End Workflow

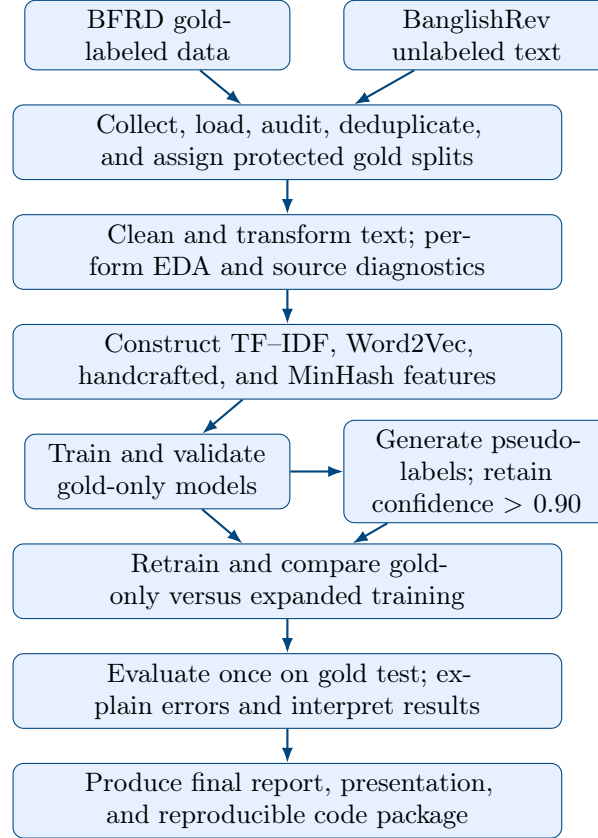


Figure 1: Proposed data-analytics and modeling workflow

9.2 Leakage-Controlled Experimental Procedure

The sequence in Figure 1 will be implemented as reproducible Spark pipelines. All token vocabularies, IDF statistics, scalars, Word2Vec models, and classifiers will be fitted without access to gold test records. Model selection will use only training and validation data. After the final configuration is chosen, the test partition will be scored once for the reported comparison.

The tuned gold-only Logistic Regression model will generate probabilities for BanglishRev. A record will be accepted when

$$\max\{P(y = 0 \mid x), P(y = 1 \mid x)\} > 0.90.$$

Its predicted class will become the pseudo-label, while the confidence and provenance will remain audit fields rather than predictive inputs. Acceptance counts and class balance will be examined before retraining. If the pseudo-labeled pool overwhelms the gold data or one predicted class dominates, deterministic caps and lower sample weights for pseudo-labeled rows will preserve the influence of verified supervision. A 0.95 threshold sensitivity check will be attempted if resources permit.

9.3 Explainability and Result Interpretation

Logistic Regression coefficients will be mapped back to vocabulary and engineered features. Random Forest and GBT feature importance will provide a complementary global view. LIME will be applied to approximately 10–15 representative correct and incorrect test predictions to identify locally influential words [12]. These explanations will describe model behavior, not prove a reviewer’s intent. Final interpretation will connect metric changes, feature ablations, error groups, and pseudo-label diagnostics rather than relying on a single accuracy figure.

9.4 Implementation Schedule

Table 5: Proposed six-week schedule

Week	Focus	Checkpoint
1	Acquisition, audit, deduplication, gold split	Reconciled counts, schema, and leakage-safe partitions.
2	Cleaning, tokenization, EDA, handcrafted signals	Tested preprocessing and analytical summaries.
3	TF-IDF, Word2Vec, MinHash LSH	Reusable feature pipelines and duplicate features.
4	Gold-only training and model selection	Five-model validation comparison.
5	Pseudo-labeling, retraining, ablation	Confidence audit and semi-supervised comparison.
6	Test evaluation, explanation, reporting	Locked results, report, slides, and code package.

10 Tools and Technologies

Table 6: Proposed technical stack

Category	Tool	Planned role
Programming language	Python	Pipeline orchestration, UDFs, evaluation, and visualization.
Big-data framework	Apache Spark / PySpark	Distributed ingestion, transformations, caching, and execution.
Query and audit layer	Spark SQL	Validation, EDA aggregations, provenance, and result tables.
Machine learning	Spark MLlib	TF-IDF, Word2Vec, MinHashLSH, classifiers, tuning, and evaluators [10, 11].
Bangla NLP	BNLP Toolkit or Indic NLP Library	Bangla-aware tokenization and language-specific resources.
Local interpretation	scikit-learn and LIME	Small-sample surrogate explanation and optional calibration analysis.
Visualization	Matplotlib and Seaborn	Charts, confusion matrices, and diagnostic plots.
Data access/storage	Hugging Face, CSV/JSON, Parquet	Source acquisition and efficient intermediate storage.
Environment	Google Colab, Databricks, or local Spark	Reproducible execution according to available resources.
Version control	Git and GitHub	Code history, collaboration, and release packaging.

Package versions, random seeds, configuration values, and execution instructions will be recorded in the repository. The core deliverable will not depend on an optional transformer or graph-processing package.

11 Expected Outcomes

The project is expected to produce:

- a cleaned, structured, provenance-aware dataset with documented row counts and quality checks;
- descriptive visualizations of class imbalance, review length, code mixing, promotional signals, and duplicate structure;
- reusable TF-IDF, Word2Vec, handcrafted, and near-duplicate feature pipelines;
- a trained and evaluated comparison of five Spark MLlib classifiers;
- a controlled measurement of whether high-confidence BanglishRev pseudo-labels help or harm performance;
- interpretable global feature summaries and selected LIME explanations;
- practical recommendations about useful features, confidence filtering, and model limitations; and
- a reproducible code repository, final report, presentation, and demonstration.

A successful academic outcome will not require pseudo-labeling to improve performance. If the expanded condition performs worse, the documented result will still provide meaningful evidence about domain shift and confirmation error. No numerical model outcome is claimed in this proposal before the experiments are conducted.

12 Project Scope

12.1 Included

The core scope includes the BFRD gold benchmark, the textual portion of BanglishRev, Spark-based auditing and EDA, Bangla-aware preprocessing, four feature groups, five classical classifiers, one round of confidence-filtered pseudo-labeling, gold-test evaluation, feature ablation, error analysis, and lightweight explanations. The work is designed for completion within a single semester and approximately six implementation weeks.

12.2 Excluded or Optional

The core project will not collect private reviews directly from users, infer reviewer identity, construct reviewer-social networks, use ratings as fake-review labels, evaluate on pseudo-labeled records, or deploy an autonomous moderation system. BanglishRev images will not be processed. Fine-tuned BanglaBERT, iterative pseudo-labeling, GraphFrames, full SHAP analysis, probability calibration beyond diagnostics, and a Streamlit/Gradio interface are optional stretch goals. They will be attempted only after all core comparisons are complete.

The resulting model will be a research prototype for educational analysis. It will not be represented as a universal detector for every Bangla domain or as proof that an individual reviewer acted deceptively.

13 Ethical and Privacy Considerations

The selected datasets are publicly accessible research resources, but public availability does not remove the obligation to minimize harm. The project will follow repository licenses and attribution requirements, retain only fields necessary for the analysis, and avoid redistributing raw text or associated images beyond permitted terms. Any names, addresses, telephone numbers, account identifiers, or other personally identifying content encountered during ingestion will be excluded from analytical outputs and demonstrations.

Pseudo-labels will be clearly identified as model-generated rather than verified facts. A prediction of “fake” can be wrong and may cause reputational harm if treated as an accusation. Consequently, the proposed system will be described as decision support for prioritizing human review, not as an automatic enforcement mechanism. Examples in the report will be minimized or paraphrased where necessary, and aggregate statistics will be preferred.

Bias may arise from BFRD’s food/social-media domain, class imbalance, linguistic variation, and uneven representation of code-mixed styles. The gold-test design, source-aware diagnostics, class-wise metrics, and error analysis will make these limitations visible. Model explanations will be interpreted as evidence about learned correlations rather than evidence about a reviewer’s intention.

14 Work Distribution

Table 7: Proposed responsibility allocation

Team member	Student ID	Primary responsibilities
Syed Abir Hossain	20220104013	Model training/tuning, pseudo-labeling, evaluation, explainability, and pipeline integration.
Alim Bin Yeasin	20220104016	Bangla preprocessing, TF-IDF/Word2Vec pipelines and MinHash near-duplicate features.
Shuhrid Abrar	20220104028	Dataset acquisition, explicit schemas, Spark SQL quality checks, gold split validation, and exploratory data analysis.

All members will jointly review experimental validity, maintain shared documentation, verify results, write the final report, prepare slides, and rehearse the demonstration. Primary ownership indicates coordination responsibility rather than exclusive authorship; code review and reproducibility checks will cross individual task boundaries.

15 Expected Challenges

Table 8: Anticipated challenges and mitigation measures

Challenge	Planned response
Strong class imbalance in BFRD	Use instance weights where supported; prioritize macro-F1, PR-AUC, and fake-class recall over accuracy.
Domain shift between food/social-media and e-commerce reviews	Preserve source provenance, use BanglishRev only for training, and judge all final claims on the gold BFRD test set.
Noisy or overconfident pseudo-labels	Require confidence > 0.90 , audit calibration and class balance, lower pseudo-label weights, cap volume, and optionally test 0.95.
Bangla spelling variation and code mixing	Apply Unicode normalization, Bangla-aware tokenization, code-mix measurements, and conservative cleaning.
Duplicate leakage	Detect exact/near duplicates and keep duplicate groups within one partition.
High-dimensional or incompatible features	Use sparse TF-IDF for linear models, compact Word2Vec/engineered vectors for trees, and non-negative inputs for Naive Bayes.
Large BanglishRev size and similarity cost	Exclude images, use Parquet/caching/partitioning, approximate MinHash search, and sample only when resource limits require it.
Difficulty interpreting predictions	Combine global coefficients/importances with targeted LIME explanations and structured error analysis.
PySpark environment or dependency issues	Pin versions, test installation early, document setup, and retain Co-lab/Databricks/local Spark alternatives.

The team will record deviations from the proposed plan, including sampling or feature reductions caused by resource limits. This will distinguish methodological decisions from unreported implementation constraints.

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